

Rishi Natarajan, Adam Inga, Rachel Zheng
STAT 3355 Project Proposal

Dataset Information:

Dallas Police Incidents includes date, time, location, and victim information, weapon, gang affiliation, and type of crime committed (**Our Dataset**):

https://www.dallasopendata.com/Public-Safety/Police-Incidents/qv6i-rri7/about_data

Crude Oil Prices: West Texas Intermediate (WTI) (**Our Dataset**)

Gives the Crude oil prices every day per barrel of oil.

<https://fred.stlouisfed.org/series/DCOILWTICO#>

Dallas Police Arrests includes the date, time, location, and arrestee of a crime arrest (**Required Dataset**):

<https://www.kaggle.com/datasets/aaronnichowilliams/dallas-police-activity>

Background and Motivation / Rationale:

We would like to look at Dallas Crimes that are committed within the years 2020-2026. The main issue we have with the Dallas Police Arrests Dataset is that it does not have recent information, but we can get the basic information such as the **arrest city, arrest day, weapon used, whether it is drug related, etc.** However, the Dallas Police Incidents dataset gives us the recent data we are missing. It also has additional columns we can use such as **when the crime started (Date1 of Occurrence)** and **when the crime ended (Date2 of Occurrence), Type Location (Apartment, Single-Family Residence, Financial Institution), and whether there is gang affiliation.** We would also like to use a crude oil dataset because this can help tell us the state of the economy better rather than tracking the S&P 500 for example. This information would be extremely useful, because we can use this data to observe patterns in where crimes are most likely going to be located, and this can help alert DFW about high risk areas and send adequate police towards those high risk areas.

Potential Questions:

- At what ages are crimes mostly committed?
- What time during the day are different types of crimes most likely to be committed?
- **Are crimes more likely to be committed during economically unstable periods? (Use Crude oil dataset to prove when economically unstable periods are).**
- Are most crimes gang affiliated or individuals?
- Which areas are experiencing more/less crime in Dallas?
- Which ethnicity is most likely to commit and be a victim of a crime?
- Is there a season pattern in crime?
- How does gender relate to crime involvement?
- Are late night crimes more likely to involve the use of weapons?
- Do different types of crime tend to occur in specific areas?
- Are day or night crimes more violent?

We chose these questions because they can help us identify patterns regarding who, where, why (Gang Affiliation or Individual), and when these crimes are being committed (bad economic period and time of day). Answering these questions can help us understand why and where these crimes are occurring, and if the city of Dallas has this information, they can use it to de-incentivise crime and support the people there in the ways where they need help. We also are using a crude oil dataset to answer one of our key questions whether crimes occur during economically unstable periods, since oil prices drag the economy if it spikes or drops. We want to see if this is a correlation or a causation.